

Evolutionary Design of Combinational Digital Circuits: State of the Art, Main Problems, and Future Trends

Adam Slowik¹ and Michal Bialko²

Koszalin University of Technology

Department of Electronics and Computer Science

Sniadeckich 2 Street, 75-453 Koszalin, POLAND

¹)aslowik@ie.tu.koszalin.pl ²)bialkomi@ie.tu.koszalin.pl

Abstract

In this paper a state of the art, main problems and future trends concerning evolutionary design of combinational digital circuits are presented. Several evolutionary algorithms for design and optimization of combinational digital circuits are also shortly described. In evolutionary design methods the main problem is related to "combinatorial explosion" existing in those methods. Because evolutionary design methods are based on "generate and test" model, therefore together with growing of digital circuit input number, the number of potential combinations which must be tested are growing exponentially. This problem is a "bottle-neck" of those methods, and therefore it is discussed in details in this paper. To solve this problem it becomes important to elaborate efficient decomposition techniques of the designed circuits to some less complex sub-circuits, and then to design each of them independently. Also, it is important to create novel genetic operators (as mutation and crossover) which will always lead to acceptable solutions (solutions which satisfy the truth table). Due to such kind of operators, any repair mechanisms need not necessarily be used, and therefore the "generate and test" model will be simplified only to generation of new solutions, and therefore evolutionary design methods will become more effective.

1. Introduction

Digital circuits are divided into combinational circuits in which the state of inputs determine the state of outputs unambiguously, and sequential circuits in which the state of outputs is dependent on the state of inputs and previous states of circuits. The voltage logic gates which can be included to the set of elementary combinational circuits are mostly used to physical realization of combinational digital circuits. These gates possess many inputs and one out-

put. More complicated combinational circuits such as: multiplexers, demultiplexers, adders, which have wide practical application in currently produced electronic hardware can be build of simple logic gates. Minimization of power consumption and minimization of area occupied by circuit becomes an important problem in complex electronic circuits. Therefore, the design of circuits fulfilling prescribed requirements (given by the truth table) and having minimal number of gates necessary to its physical realization is very important. Up to the present time majority of electronic systems of any level of complexity were created by specialized designers - experts using design heuristics. Designers possess knowledge on operation of each elementary sub-circuits, and know the designing rules. Due to this abilities, more complex electronic systems consisting of elementary parts can be created by designers [1]. Such design method is obviously limited by experience and designer knowledge. It is also possible to say, that final product such as the designed electronic circuit will be as good as the designer knowledge and his experience permits [1].

Among elaborated combinational digital circuit design methods we can distinguish two the most popular: Karnaugh Map method [2], and Quine-McCluskey method [3, 4]. During the design of combinational digital circuits based on those methods usually only NOT, AND, and OR gate types are used. Digital circuits design with XOR gates is more complicated and Karnaugh Map method and Quine-McCluskey method do not support design with XOR gates [5]. XOR gates are used in arithmetic circuits and telecommunication circuits especially. The logical circuits designed using XOR gates can consist smaller number of gates, and smaller number of connections between them; due to this property the designed circuits can occupy smaller chip area during its realization in VLSI technology. Such circuits can also be composed of smaller number of cells in the case of their realization in FPGA (Field Programmable Gate Arrays) circuits. Many methods have been developed to de-

sign digital circuits using NOT, AND, OR, and XOR gates. Among them, we can mention the method [6] in which an extension of Quine-McCluskey method allowing application of XOR gates has been proposed, and the method [7] where heuristic algorithm to the design of digital circuits using NOT, AND, OR, and XOR gates has been shown.

Among programs used for synthesis of logic circuits we can mention: ESPRESSO, and SIS. System ESPRESSO [8] is a complex heuristic program of boolean functions minimization developed at the beginning of eighty years at the University of California, Berkeley. Due to its efficiency, this program is recognized as a model program for optimization of logical expression, despite of many years passing from its development. System SIS [9] (also elaborated at Berkeley) is composed of the set of procedures for design and optimization of combinational and sequential digital circuits in binary logic. It includes the procedures which are available in ESPRESSO, and additionally allows for realization of logical circuits using gates from predefined gate library.

Currently, evolutionary algorithms to the design of combinational digital circuits are used more frequently [10, 11, 12]. In evolutionary algorithms the process of new circuits creation is fundamentally different from traditional design process.

In evolutionary circuits creation, the designer's knowledge and experience are replaced by evolution process. Evolutionary design of digital circuits has less constraints than design based on designer's knowledge and experience [14]. Designers are not only limited by technology in which circuit will be produced, but also by their own habits (routine), imagination, and creative thinking. However, the evolutionary method application to the design of digital circuits can go out of these constraints and to obtain access to new possibilities [14].

In this paper an actual state of knowledge and future trends in evolutionary design of combinational digital circuits is presented. The existing methods based on evolutionary information processing, used to design of combinational digital circuits are described and also, the main problems related to evolutionary design are pointed out and discussed.

2. State of the Art of Evolutionary Design of Combinational Digital Circuits

One of the first works where genetic algorithms to design combinational digital circuits were used is Louis doctoral dissertation in 1993 [17]. In this dissertation a connection of knowledge-based system with genetic algorithm to design digital circuits is presented. Also, in this paper the concept of gate pattern (see Figure 1) as a two dimensional array of possible gates and connection between them is presented for the first time. This concept of gate pattern was a big

step in efficiency increasing of genetic algorithms as design tools.

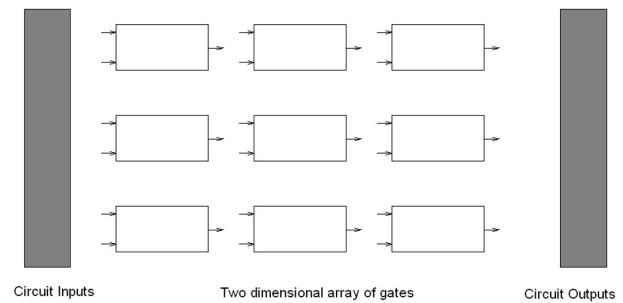


Figure 1. Concept of gate pattern [17]

In 1992 Koza [18] used genetic programming to design combinational digital circuits [18]. In his paper, he designed a two-bit adder using the gates: AND, OR, NOT. However, his research was only limited to creation of functional circuits, but not to optimization of them (similarly as in Louis dissertation [17]). In first papers on combinational digital circuits design using genetic algorithms, the binary representation of individuals (BGA algorithm - Binary Genetic Algorithm) was applied. This representation was not effective with respect to strong growth of chromosome length together with growth of task dimensionality; therefore other representations having values in determined number system (for example: senary, octal, hexadecimal) were used to evolutionary circuit design. In 1996, in paper [19] Coello proposed genetic algorithm with variable dimensionality of potential solutions coding; this proposition is connected to application of various number systems to solution coding. In proposed algorithm, it was assumed that the kind of number system is dependent on the size of gate pattern. This algorithm has been named NGA (N-cardinality Genetic Algorithm). Combinational digital circuits designed using NGA algorithm have been compared with circuits designed using BGA algorithm, and it was shown that results obtained using NGA algorithm are better than using BGA algorithm. In 1997, in paper [1] Miller proposed an approach similar to Coello approach from paper [19]. In proposed algorithm Miller used more compact representation of individual (in relation to BGA algorithm) based on integer numbers. Results obtained using Coello method [19], and Miller method [1] are comparable. In 1998, in paper [20] Kalganova proposed extension of evolutionary algorithm application to design of combinational digital circuits with multi-valued logic. In this paper the set of logical gates used in circuit design was increased, and the three-valued one-bit adder was designed by proposed method. This article was probably the first in which an application of algorithm based on evolutionary techniques to design of multi-valued logic com-

binational digital circuits was used. In 1999, in paper [21] Miller shown also the application of evolutionary algorithm as a tool for digital circuits design. In this paper a multiplier circuit multiplying two three-bit numbers designed using evolutionary method has been presented. In comparison to test circuits designed using evolutionary methods (i.e. benchmark circuits used in evaluation of efficiency of evolutionary algorithms) presented in other papers, the circuit of paper [21] was the most complex. Also in 1999, in paper [22] Kalganova has proposed the evolutionary algorithm with multi-objective fitness function to design combinational digital circuits. The main goal of the presented method was a simultaneous optimization of circuit functionality together with the structures of its gates. In 2000, in paper [23] Coello has proposed a multi-objective version of evolutionary algorithm to digital circuit design. The main idea of his MGA algorithm [23] (Multiobjective Genetic Algorithm) was an utilization of multi-objective optimization (similarly as in VEGA [24] system) to solve optimization problem consisting of k equality constraints. The k value represents the number of rows in the truth table of the designed circuit. MGA algorithm was operating on population consisting of $k+1$ sub-populations, where each subpopulation was optimizing one constraint. The experiments performed in paper [23] have been proved that using MGA algorithm it is possible to obtain circuits having smaller number of gates than circuits designed by human designers using Karnaugh Map method and Quine-McCluskey method or using earlier elaborated NGA algorithm. Also, in 2000 in paper [28] Vassilev and Miller have presented the four-bit multiplier (circuit multiplying two four-bit numbers) designed using evolutionary techniques. The evolutionary design of such circuit is a very complicated task; to design this circuit it was required 700.000.000 algorithm generations. This circuit consists of 57 two-inputs gates (35 AND gates, 22 XOR gates) and 10 NOT operations. This circuit has lower number of gates (by 11%) compared to the circuit designed using traditional methods.

In 2003 in paper [25] an application of evolutionary algorithm to design and optimization of combinational digital circuits with respect to minimal number of transistors in designed circuit has been presented. It was shown that changing the optimization criterion it is possible to obtain the circuit solutions having smaller area in silicon chip of VLSI circuit. In 2004, in paper [26] Slowik has presented evolutionary algorithm with multi-layer chromosome to design combinational digital circuits. It was shown that changing the circuit representation in algorithm from single-layer chromosome to multi-layer chromosome (see Figure 3b) can bring rational advantages. Algorithm described in paper [26] has been named MLCEA (Multi-Layer Chromosome Evolutionary Algorithm). The experiments taken in paper [26] have been proved that algorithm MLCEA is more ef-

fective than MGA and NGA algorithms. In 2005 in paper [29] Coello has shown an application of evolutionary algorithm together with case-based reasoning to design and optimization of combinational digital circuits, and a new design rule using XOR gates has been shown too. In 2006 in paper [27] a modification of MLCEA algorithm to design digital circuits with respect to minimal number of transistors using multi-layer chromosomes was presented. The circuits obtained using method from paper [27] were composed of smaller number of transistors than circuits described in paper [25].

3. General Concept of Combinational Digital Circuits Design Methods Using Evolutionary Algorithms

The general concept connected to the design of combinational digital circuits in majority papers (besides paper [18]) described in section two is related to so called pattern of gates. The size of the gate pattern is determined at the start of evolutionary algorithm. In Figure 2 the pattern of gates with the size equal to $t \cdot t$ is shown (this size may be not always quadratic). In each pattern $t \cdot t$ there exist places for potential circuit gates. The pattern also consists of k inputs, and f outputs, which represent circuit inputs and circuit outputs. The main goal of each evolutionary algorithm used to design and optimization of combinational digital circuits is determination a such set of gates in the pattern, and such set of connections between gates, that the designed circuit fulfills assumed truth table. Also, depending on assumed optimization criterion, it is required to design the circuit consisting of minimal number of gates, minimal number of transistors, or circuit having minimal time of signal propagation from its input to its output.

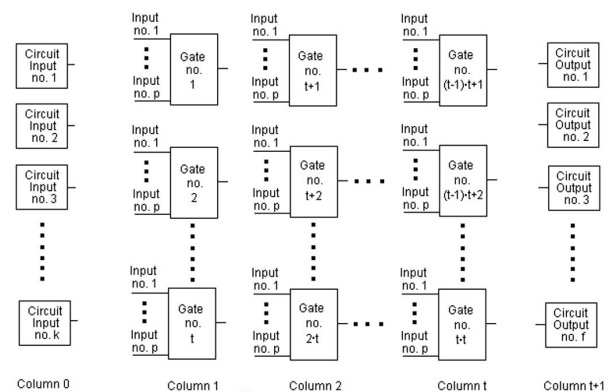


Figure 2. Pattern of designed digital circuit

The pattern of gates shown in Figure 2 is coded in algorithm as a single-layer chromosome or as a multi-layer chro-

mosome (depending on the type of evolutionary algorithm chosen to design). In Figure 3a single-layer chromosome is presented, and in Figure 3b the multi-layer chromosome is shown, and in both chromosomes the pattern of Figure 2 is coded.

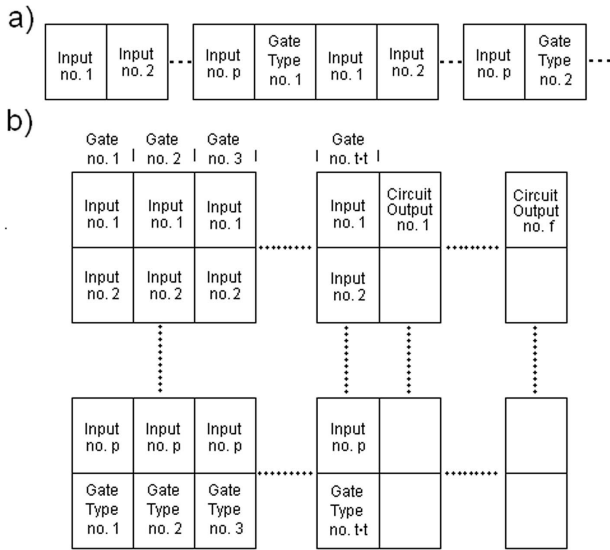


Figure 3. Pattern of gates coded in the form of single-layer chromosome (a), and multi-layer chromosome (b)

In the case of pattern of Figure 2 coded as a multi-layer chromosome (Figure 3b), each gate is represented by a single column of chromosome; therefore, during cross-over operation the whole gates without damage of their internal structure can be moved inside pattern. In the case of pattern coded as a single-layer chromosome (Figure 3a) the successive values are placed in single numeric sequence. Hence, due to single-layer structure of the chromosome, it can happen that during the cross-over operation a non-acceptable circuit will be generated, for example: NOT gate having two inputs, or XOR gate without one input connection, etc. In order to prevent this situation, it is required to apply repair mechanisms or to introduce so called prohibited cut points of chromosome during the cross-over operation. In the case of multi-layer chromosomes, the generated new circuits after cross-over operation are always acceptable. In Table 1, the exemplary truth table of designed circuit is presented. This circuit possess three inputs (X, Y, Z), and one output (F), and can be practically realized as in Figure 4a.

The circuit from Figure 4a is represented by the pattern of gates shown in Figure 4b. For pattern of Figure 4b the value of parameter t is equal to 5. In this pattern the 25 gates can be located. The gates forming the designed circuit are marked by grey color. The symbol "DC" represents

the direct connection of first input of a given gate with its output. The pattern from Figure 4b, can be represented by the multi-layer chromosome shown in Figure 4c.

Table 1. Truth table of exemplary circuit

X	Y	Z	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

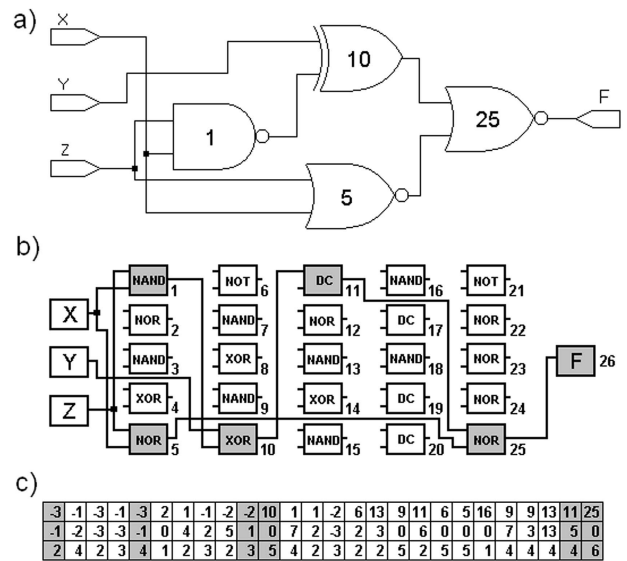


Figure 4. Exemplary digital circuit fulfilling the truth table from Table 1 (a), pattern of gates representing this circuit (b), multi-layer chromosome representing this pattern (c)

The successive circuit inputs are represented by negative numbers ("1" - X; "2" - Y, "3" - Z). The numbers placed in the third layer of the multi-layer chromosome represent respectively: 1 - NOT gate; 2 - NAND gate; 3 - XOR gate; 4 - NOR gate; 5 - DC - direct connection of first input of a given gate with its output, 6 - circuit output, 0 - lack of connection of a given input. Of course the overall information processing in the algorithm is based on rules existing in evolutionary algorithm. The cross-over operations depend on cutting of two randomly chosen individuals (chromosomes) through all layers in a single (randomly chosen)

point, and then the cut parts of two randomly selected chromosomes are exchanged, and the mutation operations depend on changing values for particular genes; these changes correspond to changing the gate connections in the pattern, or changing the gate types.

4. Main Problems and Future Trends in Evolutionary Circuits Design

Computational complexity of evolutionary methods used for combinational digital circuit design is higher than complexity of heuristics in SIS system. It is caused by fact, that evolutionary circuit design is based on "generate and test" schema. According to this schema, when we have binary signals 0, 1, and the circuit having for example 6 inputs, and 4 outputs, then we have to check its operation based on $4 \cdot 2^6 = 256$ output values following from this circuit truth table. Assuming that algorithm population consists of 100 individuals, then the 25.600 checks of correctness of given circuit operation must be performed in each algorithm generation since the evolutionary circuits design is based on pattern of gates, and in the case when the gate pattern includes 25 elements, and using 5 types of gates (NOT, AND, OR, XOR, DC) a total number of combinations of potential gate arrangements in the pattern is approximately $5^{25} \approx 2.98 \cdot 10^{17}$. Due to this huge number of combinations, the searching of the solution space is very time-consuming. This problem is named in literature as a scalability problem i. e. the very strong increase of potential solution space together with increase the number of circuit inputs. In order to increase the effectiveness of evolutionary techniques of combinational circuits design it is possible to apply following improvements: algorithm initialization by an initial solution obtained from the SIS system (or other), decompose the circuit into n simpler sub-circuits, and their successive design, using more complex base elements as for example the replacement standard two-input gates replacing by n -input modules realizing described more complex logic functions. The future research can concentrate on the attempt of elaboration of genetic operators guaranteeing that the designed circuit will be still functional circuit, that is the circuit fulfilling the truth table, after using these genetic operators. Having such genetic operators, and initializing the population by initial solution fulfilling the given truth table, obtained using for example from SIS program, we can expect considerable increasing of efficiency of combinational digital circuits design, and optimization methods, based on evolutionary algorithms. The increase of efficiency will be caused by elimination of testing each potential circuit in the population with respect to fulfilling of limitations following from its truth table. Then model of the type "generate and test" will be simplified to the model of the type "generate", what will assure more effective design process. Also, in

the future works, it is recommended to study the possibility of application of other computational intelligence algorithms to design combinational digital circuits, and comparison the efficiency of these algorithms between them. Such algorithms as: particle swarm being a model of behaviors of birds flocks (fish, bees, and the like), ant colony optimization simulating behaviors of ant colony, differential evolution, and artificial immune systems simulating the operation of human immunological system may be considered in the future papers. The first papers referring to design of combinational digital circuits using computational intelligence algorithms other than evolutionary algorithms already arisen, but this subject is still only slightly investigated. Publications describing comparative results between computational intelligence techniques, and their efficiency in application to design and optimization of combinational digital circuits, do not exist. In spite of the large number of problems, which are still open to solving in the area of evolutionary design of combinational digital circuits, the huge advantage of these methods is the possibility of complete automation of the design process. The user needs not to know the rules referring to combinational circuit design, and needs not to know the standard design methods such as: Karnaugh Map method, Quine-McCluskey algorithm, and the heuristics from the SIS system. Also, in the case of design methods based on evolutionary algorithms, it is possible to use components of any type: XOR, NXOR, NOT, OR, NOR, ect.

5. Conclusion

In the paper, evolutionary methods of combinational digital circuits design are presented. The general concept of application of evolutionary algorithms to this kind of problems are discussed, and main problems connected to evolutionary design of combinational digital circuits are shown and together with potential trends of future research are considered.

References

- [1] J.F. Miller, P. Thomson, T. Fogarty, "Designing Electronic Circuits Using Evolutionary Algorithms. Arithmetic Circuits: A Case Study", Genetic Algorithms and Evolution Strategy in Engineering and Computer Science, pages 105-131, Wiley, Chichester, England, 1997
- [2] M. Karnaugh, "A Map Method for Synthesis of Combinational Logic Circuits", Transaction of the AIEE, Communications and Electronic, 72(I):593-599, November, 1953
- [3] W. V. Quine, "A Way to Simplify Truth Function", American Mathematical Monthly, 62(9):627-631, 1955

- [4] E. J. McCluskey, "Minimization of Boolean Function", Bell Systems Technical Journal, 35 (5):1417-1444, 1956
- [5] C. A. Coello, A. D. Christiansen, A. H. Aguirre, "Use of Evolutionary Techniques to Automate the Design of Combinational Circuits", International Journal of Smart Engineering System Design, 2000
- [6] B. C. H. Turton, "Extending Quine-McCluskey for Exclusive-OR Logic Synthesis", IEEE Transaction on Education, No. 1, pp. 81-85, 1996
- [7] D. Debnath, T. Sasao, "A heuristic algorithm to design AND-OR-EXOR three-level networks", in Proceedings of Asia and South Pacific Design Automation Conference, pp. 69-74, Feb. 1998
- [8] G. De Micheli, "Synthesis and Optimization of Digital Circuits", WNT, Warszawa, 1998, (in Polish)
- [9] T. Luba, "Synthesis of Digital Circuits", WKL, Warszawa, 2003, (in Polish)
- [10] Arabas J. "Lectures on Evolutionary Algorithms", WNT, Warszawa, 2001, (in Polish)
- [11] Goldberg D. E.: Genetic Algorithms in Search, Optimization, and Machine Learning, Addison Wesley Publishing Company Inc., 1989.
- [12] Michalewicz Z.: Genetic Algorithms + Data Structures = Evolution Programs, Springer-Verlag, 1992.
- [13] Slowik A., "Design and Optimization of Digital Electronics Circuits Using Evolutionary Algorithms", Doctoral Dissertation, Koszalin University of Technology, Department of Electronics and Computer Science, March 2007, (in Polish)
- [14] Greene J., Simulated Evolution and Adaptive Search in Engineering Design -Experiences at the University of Cape Town, in 2nd Online Workshop on Soft Computing, 1997
- [15] Miller J. F., Job D., Vassilev V. K., "Principles in the Evolutionary Design of Digital Circuits - Part I", Genetic Programming and Evolvable Machines, 1(1/2), pp. 7-35, 2000
- [16] Koza J., Keane M., Streeter M., "Evolving Inventions", Scientific American Magazine, February 2003
- [17] Louis, Sushil J., "Genetic Algorithms as a Computational Tool for Design", Ph. D. Dissertation, Department of Computer Science, Indiana University, 1993
- [18] Koza, John R., "Genetic Programming. On the Programming of Computers by Means of Natural Selection", The Mit Press, 1992
- [19] Coello Coello, Carlos A.; Christiansen, Alan D. and Hernandez Aguirre, Using Genetic Algorithms to Design Combinational Logic Circuits, ANNIE'96. Intelligent Engineering through Artificial Neural Networks, Volume 6. Smart Engineering Systems: Neural Networks, Fuzzy Logic and Evolutionary Programming. Edited by: Cihan H. Dagli, Metin Akay, C. L. Philip Chen, Benito R. Fernandez and Joydeep Ghosh, pp. 391-396. November, 1996
- [20] T. Kalganova, J. Miller, T. Fogarty, "Some Aspects of the Evolvable Hardware for Multiple-Valued Combinational Circuit Design", Proceedings of the Second International Conference on Evolvable Systems (ICES'98), pages 78-89, Lausanne, Switzerland, 1998
- [21] J. Miller, T. Kalganova, N. Lipnitskaya, D. Job, "The Genetic Algorithm as a Discovery Engine: Strange Circuits and New Principles", In Proc. of the AISB Symposium on Creative Evolutionary Systems (CES'99), Edinburgh, UK, 1999
- [22] T. Kalganova and J. Miller., "Evolving more efficient digital circuits by allowing circuit layout and multi-objective fitness", Proc. of the First NASA/DoD Workshop on Evolvable Hardware, pp. 54-63, Los Alamitos, California, 1999
- [23] Carlos A. Coello Coello, Arturo Hernandez Aguirre, and Bill P. Buckles, "Evolutionary Multiobjective Design of Combinational Logic Circuits", Proceedings of the Second NASA/DoD Workshop on Evolvable Hardware, pages 161-170, Los Alamitos, California, July 2000
- [24] J. D. Schaffer, "Multiple Objective Optimization with Vector Evaluated Genetic Algorithms", In Genetic Algorithms and their Applications: Proceedings of the First International Conference on Genetic Algorithms, pages 93-100, Lawrence Erlbaum, 1985
- [25] P. Nilagupta and N. Ou-thong, "Logic function minimization base on transistor count using genetic algorithm", Proc. 3rd Information and Computer Engineering Postgraduate Workshop, Songkla, Thailand, 2003
- [26] Slowik A., Bialko M.: Design and Optimization of Combinational Digital Circuits Using Modified Evolutionary Algorithm, Proceedings of Seventh International Conference on Artificial Intelligence and Soft Computing ICAISC 2004, Lecture Notes in Artificial Intelligence, Volume 3070/2004, pp. 468-473, 2004
- [27] Slowik A., Bialko M.: Evolutionary Design and Optimization of Combinational Digital Circuits with Respect to Transistor Count, Bulletin of the Polish Academy of Sciences, Technical Sciences, Volume 54, Number 4, pp. 437-442, 2006
- [28] V. Vassilev, D. Job, and J. Miller. Towards the automatic design of more efficient digital circuits. In J. Lohn, A. Stoica, and D. Keymeulen, editors, The Second NASA/DoD workshop on Evolvable Hardware, pages 151-160, Palo Alto, California, 13-15 2000. IEEE Computer Society
- [29] Eduardo Islas Prez, Carlos A. Coello Coello, and Arturo Hernandez Aguirre, Extraction and Reuse of Design Patterns from Genetic Algorithms using Case-Based Reasoning, Soft Computing—A Fusion of Foundations, Methodologies and Applications, Springer-Verlag, Vol. 9, No. 1, pp. 44-53, January 2005